EL NIÑO ACTION BRIEFING · 10 OCTOBER 2026 · VERIFIED RUN

ZIP 94086, Sunnyvale, California

A strong El Niño raises the odds of a wet winter in central Sunnyvale. The deaths on the record here are not from floods or power cuts: they are cold, wet nights for about 330 people living in vehicles and outdoors, on streets where a new rule now makes RVs move every 72 hours. The most important action is a winter motel program renewed before December, with the parking rule paused on storm days.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
The Climate Impact Lab region containing Sunnyvale, Santa Clara County (name not resolved)	None published for the US, California, or the county	Jun 2026 – Feb 2027 (heat only; not this hazard)	Cold and wet exposure of unhoused people; flood-channel overtopping; storm power outages; generator carbon monoxide

22/22 recommendations verified. 1 confirmed, 10 corrected (partially confirmed), 1 unverified, 10 contraindicated. The funding target fell from the panel's $289k to about $139k, down 52%: the pilot's room rate included a 20% contingency, the city already has a draft $27,000-a-year line for the same program, rooms were cut from 30 to 20, and an unsupported $25k storm-loss fund was dropped. The El Niño premise was tested, not assumed: we classed 64 winters of daily rainfall at Moffett Field, about 3 km from the ZIP,

* Suggested formulation: check with a health worker before sharing.

EXECUTIVE SUMMARY — AT A GLANCE



37.37°N 122.02°W, United States of America · Natural Earth

- **The risk.** Deaths on the record here come from cold, wet nights for about 330 people living in vehicles and outdoors, on streets where a new rule makes RVs move every 72 hours.
- **First action.** The City of Sunnyvale and the county renew and enlarge the winter motel program by 1 December, released on the county's bad-weather declaration, with pets, partners, and belongings allowed.
- **The money.** About $54k floor, $139k target, $383k ceiling; how much is needed depends on how many county bad-weather nights this winter brings.
- **Do not.** Do not tow RVs, or clear people from the channels, in storm season without somewhere legal to go.

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by the official El Niño index. No Climate Impact Lab figure covers this place or this hazard; the record carries it instead. Method caveat: the City of Sunnyvale's website and the county's news pages refused automated reads, so some city facts rest on page extracts cross-checked against local press, and are marked where they do.

DOES EL NIÑO MATTER HERE? THE PREMISE, TESTED

Moffett Field daily rainfall (NOAA GHCN-Daily, USW00023244), October to March, winters 1950–51 to 2018–19, keeping the 64 winters with at least 90% of days reported. Winters are classed by the peak Oceanic Niño Index from October to February. 1997–98 is missing too many days to count. Read every difference as a tilt, not a rule.

WINTERS	OCT–MAR RAIN (MM)	DAYS WITH 25 MM OR MORE	LARGEST 3-DAY TOTAL (MM)	WINTERS WITH A 3-DAY TOTAL OF 75 MM OR MORE
Strong El Niño (6)	427	3.7	62	2 of 6
Moderate or weak El Niño (16)	293	1.9	53	1 of 16
Neutral (20)	291	1.4	53	3 of 20
La Niña (22)	279	1.4	55	4 of 22
All winters (64)	300	1.7	55	10 of 64

- **About 40% more rain, and twice the soaking days.** Strong El Niño winters averaged 427 mm against 300 mm, and 3.7 days with 25 mm or more against 1.7.
- **But not every time.** 1982–83 (593 mm) and 1972–73 (566 mm) were the two wettest winters on the record; 1965–66 (239 mm) and 2015–16 (295 mm) were ordinary. San José's shorter record (1998–2025, USW00023293) points the same way: strong winters 358 mm against 272 mm.

* Suggested formulation: check with a health worker before sharing.

- **What NOAA says now.** "El Niño continues to strengthen, with a strong-to-very strong El Niño likely through January-March 2027" (CPC, 8 October 2026), and its seasonal outlook favors a wetter-than-normal winter near the California coast.

EXPOSURE, HONESTLY

WHO IS AT RISK, AND FROM WHAT

- **People living in vehicles and outdoors.** Sunnyvale's 2025 count found 421 homeless people, 328 of them unsheltered. Two city documents put the share living in vehicles at 75% (the draft homelessness plan: 246 in vehicles, 82 outdoors) and 83.8% (the count summary, from 48 interviews). The city puts the RV count at 120–150. Hypothermia caused or contributed to **42 deaths in Santa Clara County in 2021–2023, up from 11** in the three years before (CDC data, reported by KFF Health News). The North County Shelter stopped new referrals in April 2024.
- **A new parking rule with no weather exemption.** Since 16 July 2026, RVs must move at least 1,000 feet every 72 hours, with towing as a last resort. The city judged its own safe-parking lot "not viable" in April 2026; private and faith-run safe-parking sites are allowed.
- **Older residents, a minority in a young ZIP.** Of 49,746 residents (median age 33.5), 5,670 are 65 or over and 1,488 of them report a disability. 29.6% of households are one person. Six licensed elder-care homes hold 138 beds, the largest Sunnyside Gardens on Carson Drive (84). (ACS 2024 5-year; CDSS.)
- **People whose lives run on electricity.** 162 Medicare beneficiaries use electricity-dependent equipment, 27 use home oxygen, and 28 rely on dialysis (HHS emPOWER, updated 7 October 2026). Central Sunnyvale is outside the High Fire-Threat District, so PG&E's wildfire-linked battery help does not reach it.
- **The flood channels.** The Sunnyvale East and West Channels were built for about a 10-year storm and flooded in 1963, 1968, 1983, 1986, and 1998. Most of the mapped flooding falls in 94085 and 94089, but at the "Major" stage it reaches apartments on East Evelyn Avenue inside this ZIP. Only 3 flood-insurance claims have ever been filed in 94086, $8,854 paid in all.
- **Languages and housing.** 2,581 households have limited English: 1,593 speak an Asian or Pacific Island language and 582 Spanish. 69.5% of households rent; there are 468 mobile homes.

WHAT THIS PLACE SHOULD DO

RANKED BY LIVES PER EFFORT

Actions 1 to 3 protect the people the record says winter kills here. Actions 4 to 7 protect older and powered-dependent residents. Actions 8 to

* Suggested formulation: check with a health worker before sharing.

11 cover flooding and money. Action 12 is after the peak.

1. Renew and enlarge Sunnyvale's winter motel program before December, released on the county's bad-weather declaration, with pets, partners, and belongings allowed

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY MGR · PHILANTHROPIST · ECONOMIST · SKEPTIC · FRONTLINE

ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
City of Sunnyvale (council contract with WeHOPE); County Office of Supportive Housing	Council approval by 1 December	$305 per room-night (2024–25 pilot); $95,000 top-up, see funding	Hypothermia deaths on cold, wet nights

In December 2024 the council amended WeHOPE's outreach contract to add 10 rooms at an Extended Stay America on Orleans Drive, up to 20 nights each, December to April, with two meals a day and pets allowed, budgeted at $60,987 plus a 20% contingency ($73,185 in all). The city's draft 2026–2030 homelessness plan makes the program ongoing at $27,000 a year, aiming to serve 25–45 people. The county's Inclement Weather Plan already opens extra beds on rain as well as cold: in its last published version, a low of 45°F or below with a 50% or greater chance of rain, 40°F or below otherwise, or 48 hours of rain likely to flood.

No study measures how many deaths a winter motel program prevents; the case rests on the county's hypothermia trend and on the mechanism. Whether the pilot has been renewed for 2026–27 could not be confirmed, and no 2025–26 or 2026–27 county trigger has been published. The February 2025 placement of 25 people in 18 rooms followed an encampment clearing, not a storm.

2. Pause the 72-hour RV move rule, and any towing, on every county bad-weather day through March, and name legal storm parking away from the channels north of US-101

PARTIALLY CONFIRMED | SKEPTIC · EMERGENCY MGR | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Sunnyvale City Manager and Department of Public Safety	Before the first storm	None	RV households forced to move at night in a storm, or into flood-prone streets

* Suggested formulation: check with a health worker before sharing.

The ordinance passed unanimously on 2 June 2026 and has been enforced since 16 July; it has no weather exemption. A storm-day pause is an administrative choice, and the oversized-vehicle permit streets the city is developing are the obvious place to put storm parking. Mountain View's 2022 settlement with RV dwellers guaranteed at least three miles of streets where they can park.

No study measures the effect on deaths. The argument is that the rule's cost lands exactly when the hazard does.

3. Give vehicle dwellers a battery carbon-monoxide alarm, a sleeping bag, and a tarp, with a warning in their language never to run a generator, grill, or propane heater inside

UNVERIFIED | PHILANTHROPIST · PUBLIC HEALTH · FRONTLINE · SKEPTIC | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
WeHOPE outreach team; safe-parking hosts	Handed out by 1 December	About $20,000 for 200 (estimate)	Hypothermia and carbon-monoxide deaths among those who stay in their vehicles

Most vehicle dwellers will not leave the RV that holds everything they own for a motel room, so this is the only placement that reaches them. Portable generators cause more than 70 carbon-monoxide deaths a year in the US (CPSC).*

No outcome study covers kits for vehicle dwellers, and the $100 unit cost is unsourced, so the action is unverified. It is cheap and low-risk, and California's alarm laws do not cover RVs.

4. Give an RSV shot to everyone 75 and over who has not had one, and to people 50–74 at higher risk, and a high-dose or adjuvanted flu shot to everyone 65 and over, starting in the six elder-care homes

PARTIALLY CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Care-home operators with pharmacies; Santa Clara County Public Health; Valley Health Center Sunnyvale	Before December	Covered by Medicare	Flu and RSV deaths, which outnumber storm deaths

* Suggested formulation: check with a health worker before sharing.

RSV vaccine was 75% effective against hospitalization in adults 60 and over in its first season, and 76% in those 75 and over (Surie, JAMA 2024); it fell to 58% over two seasons (JAMA 2025). It is one lifetime dose. High-dose flu vaccine gave 24.2% more protection than the standard dose in about 32,000 people 65 and over (DiazGranados, NEJM 2014, manufacturer-sponsored). The six homes hold 138 beds.*

The evidence comes from US adults and transfers well. No Sunnyvale Senior Center clinic was found for 2026, so one there would be new; outreach should run in Chinese, Vietnamese, Korean, and Spanish.

5. Enroll the 162 residents on powered medical equipment in Medical Baseline, get the 27 oxygen users backup cylinders from their supplier, and make an outage plan with each dialysis patient's center

PARTIALLY CONFIRMED | FRONTLINE · PUBLIC HEALTH · ECONOMIST | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Silicon Valley Independent Living Center (sccddar@svilc.org; 408-894-9041); medical providers	15 November	Enrollment free; lending pool $23,980	Equipment failing unnoticed in a long outage

Medical Baseline needs a medical provider's certification and gives a lower rate and extra outage alerts. It does not give backup power. PG&E's disability-access batteries are "only available during Public Safety Power Shutoffs," and central Sunnyvale is not in a fire-threat district. A small SVILC lending pool of 20 battery stations for powered equipment other than oxygen fills that gap (see funding).*

* Suggested formulation: check with a health worker before sharing.

Storm outages in Sunnyvale have been short and local (about 2,000 customers in the March 2025 storm). The 2003 New York blackout raised accidental deaths by 122% (Anderson and Bell 2012), but that was a summer outage.

6. Sunnyvale adopts and tests Valley Water's existing flood stages for the East and West Channels, with AlertSCC messages written in advance for the East Evelyn apartments and the mobile-home parks

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Sunnyvale Department of Public Safety (Office of Emergency Services); Valley Water	Tabletop test by 15 November	Staff time	A warning that exists but nobody owns, as in San José in 2017

Valley Water's July 2025 emergency action plan already sets stages at the East Channel gauge upstream of US-101: 6.5 ft Action; 8 ft Minor, when the Tasman bank overtops into the Plaza del Rey and Casa de Amigos mobile-home parks; 9 ft Moderate, "evacuations"; 10 ft Major, reaching Evelyn and Wolfe and the Windsor Ridge apartments in 94086. The plan calls itself "a Valley Water internal document" that does not override the city; no city sign-off was found.

After the February 2017 Coyote Creek flood, when about 14,000 people were ordered out, San José and Valley Water wrote a joint creek emergency plan. That is the precedent.

7. Sign up cell phones for AlertSCC door to door in apartments and mobile-home parks, in the residents' languages, and pair each older person living alone with a neighbor who checks during outages

PARTIALLY CONFIRMED | FRONTLINE · EMERGENCY MGR · SKEPTIC | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Sunnyvale OES neighborhood program and CERT (408-730-7198); park resident representatives; Sunnyvale Senior Center	Before the first atmospheric river	None	Warnings and outages that never reach isolated people

* Suggested formulation: check with a health worker before sharing.

AlertSCC reaches listed landlines automatically; cell phones and email must register. Palo Alto reported that only about a third of residents had registered (2023–24).

No trial shows check-ins reduce deaths; they are cheap and the blackout evidence is indirect. Whether AlertSCC sends in languages other than English could not be confirmed.

8. Fit carbon-monoxide alarms in apartments and mobile homes with gas appliances and no working alarm, starting with older residents

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Landlords and park owners (legal duty); CERT volunteers for older residents	1 December	Unfunded; no Sunnyvale install program found	Carbon-monoxide deaths from heaters and generators in outages

In New Mexico, alarms "may have prevented approximately half" of 136 unintentional carbon-monoxide deaths (Yoon, JAMA 1998), a modelled estimate from a case review, not a measured effect. California law already requires alarms in homes with fuel-burning appliances, and a separate state rule covers mobile homes, so the gap is compliance and dead batteries.

Target, do not blanket-buy. No local installer was found, so the cost is marked unfunded.

9. Help households apply for CARE and FERA now, and for HEAP through Sacred Heart early, before the limited money runs out

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Sunnyvale Community Services; Sacred Heart Community Service; volunteers	October and November	Paid by PG&E ratepayers (CARE, FERA) and federal LIHEAP (HEAP)	Rationed heat and power through the wet months

* Suggested formulation: check with a health worker before sharing.

CARE and FERA are PG&E discount rates, open all year with no funding cap. HEAP bill credits are federal money passed through the state and delivered in the county by Sacred Heart, which is running a "Priority Plan" because funding is limited and cannot guarantee help.

10. Renters in the East Evelyn flood area buy contents flood cover by about 1 December

PARTIALLY CONFIRMED | ECONOMIST · EMERGENCY MGR | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Residents in the mapped inundation area	About 1 December, for January cover	Premiums vary	Uninsured loss of belongings in a Major-stage flood

National Flood Insurance Program policies normally take effect 30 days after purchase. Sunnyvale's 15% discount applies only inside the mapped flood zone.

Demoted: the ZIP has had three claims in 40 years. Worth it only where Valley Water's plan shows water reaching homes.

11. Valley Water clears vegetation and inlets on the channels before the first storm, and residents collect sandbags from the Corporation Yard

PARTIALLY CONFIRMED | EMERGENCY MGR · ECONOMIST | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Valley Water; City of Sunnyvale	Sandbags from 1 November	Existing maintenance budgets; sandbags free	Channels losing capacity to regrowth and debris

* Suggested formulation: check with a health worker before sharing.

Sandbags are free and self-serve at 221 Commercial Street, 10 per address with Sunnyvale ID, from 1 November to 1 April, weekdays 7:30 am to 4 pm. Valley Water's hot-spot list names the East Channel at Evelyn Avenue, and its plan warns that fast regrowth north of US-101 worsens flooding.

12. Hold Valley Water to its 2027 permits and its 2032 completion date for the East Channel upgrade

CONFIRMED · AFTER THE PEAK | HISTORIAN | ADAPTATION INVENTORY: NOT QUERIED

OWNER	BY WHEN	COST	PREVENTS
Valley Water board; City of Sunnyvale	Permits 2027	Now $45–50M	Repeat channel flooding for 1,618 parcels

The project gives 1,618 parcels 100-year protection. It was budgeted at $38.8M from the Safe, Clean Water program; the August 2026 project list shows $45–50M with a federal loan and FEMA grant. Permits were first filed in 2017; design is due spring 2027 and completion "by 2032." An older FY29 date on the same page is stale.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	City of Sunnyvale (council, City Manager)	Sunnyvale DPS, OES & CERT	WeHOPE	Santa Clara County (Supportive Housing, Public Health)	Care homes & pharmacies	SVILC	Valley Water	SCS & Sacred Heart	Residents
1. Winter motel program	**A**		**R**	**C**					
2. Storm-day RV pause	**A**	**R**	**C**						**I**
3. Kits for vehicle dwellers	**C**		**A/R**						
4. RSV and flu shots				**R**	**A/R**				**I**
5. Medical Baseline, oxygen, dialysis				**C**		**A/R**			**R**
6. Adopt and test channel stages	**C**	**A/R**		**C**			**R**		
7. AlertSCC sign-ups and neighbor checks		**A/R**		**C**					**R**
8. CO alarms in homes		**R**							**A/R**
9. CARE, FERA, HEAP								**A/R**	**R**
10. Renters' flood cover	**I**						**C**		**A/R**
11. Channel clearing and sandbags	**R**						**A/R**		**R**
12. East Channel upgrade	**C**						**A/R**		

What the chart shows. The City of Sunnyvale is accountable for the two actions that matter most, the motel program and the storm-day RV pause, and both are decisions the council or City Manager can take this fall with no new authority. WeHOPE's five-person team does the work for actions 1 and 3, so its capacity on a storm night is the bottleneck. The Office of Emergency Services carries the warning rows, and Valley Water owns the channels but not the decision to evacuate.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

About $54k

TARGET

About $139k

CEILING

About $383k

LARGEST UNCERTAINTY

How many county bad-weather nights this winter brings, and whether the council has renewed the program

* Suggested formulation: check with a health worker before sharing.

Arithmetic. Room rate: the December 2024 pilot budgeted $60,987 for 10 rooms × 20 nights, or $305 per room-night including two meals a day and overnight coordinators, before a 20% contingency. The city's draft 2026–2030 homelessness plan already funds the program at $27,000 a year (adoption not confirmed), so that is netted off. Floor: 10 rooms × 20 nights × $305 = $61,000 − $27,000 = $34,000, + kits $20,000 = $54,000. Target: 20 rooms × 20 nights × $305 = $122,000 − $27,000 = $95,000, + kits $20,000 + batteries 20 × $1,199 = $23,980, = $138,980. Ceiling: a wet winter doubling the nights, 30 rooms × 40 nights × $305 = $366,000 − $27,000 = $339,000, + $20,000 + $23,980 = $382,980. The target fell 52% from the panel's $289k: the contingency had been counted as rate, the city line had been missed, rooms were cut from 30 to 20 because a five-person team cannot fill more on a day's notice, and a $25k storm-loss fund with no basis was dropped.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
WeHOPE, through the City of Sunnyvale contract	Action 1: winter motel rooms on county bad-weather nights	$95,000	Top-up to the city's existing program	20 rooms × 20 nights × $305 (2024–25 pilot rate) − city's $27,000; no published cost per death averted for winter shelter	Partially confirmed; contract renewal unconfirmed
WeHOPE outreach team	Action 3: kits for vehicle dwellers	$20,000	Pre-positioned stock	200 × about $100 (battery CO alarm about $22–27, plus sleeping bag and tarp; estimate)	Partially confirmed; check the alarm is rated for vehicles
Silicon Valley Independent Living Center	Action 5: small lending pool of battery stations for storm outages, for powered equipment other than oxygen	$23,980	Restricted grant	20 × $1,199 (2,048 Wh station); revenue $4.1M (year to June 2025); PG&E's program does not cover storm outages here	Partially confirmed; SVILC has not run its own pool before

The funder's one thing: most of the roughly 250 people living in vehicles will not trade the vehicle that holds everything they own for a motel room, so money beyond about 20 rooms buys empty rooms. Spend it on more nights, and on the kits, which are the only placement that reaches the people who stay put.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here, or waste the weeks before the peak.

Do not tow RVs, or clear people from the channels, in storm season without somewhere legal to go

CONTRAINDICATED SKEPTIC; ORDINANCE CONFIRMED

Sunnyvale's 72-hour rule has no weather exemption, and the city has no lot of its own. In January 2023, when San José ordered people away from its creeks, many stayed because they would not leave their belongings or had nowhere to go.

Instead: the storm-day pause and named storm parking (action 2), and outreach with secure storage before the wet season.

Do not let warming sites close, or wait to open, when a storm is forecast

CONTRAINDICATED SKEPTIC AND EMERGENCY MGR; PARTLY CONFIRMED

On 23 February 2026 Sunnyvale's notice deactivated the library warming center, citing cold, rain, and flooding (confirmed from a page extract only). In San José, a community center stayed shut next to a flooding encampment because its operator needed 48 hours' notice.

Instead: open on the forecast, 48 hours ahead, at a site that stays open through the storm (action 1).

Do not draft new flood triggers for the channels

CONTRAINDICATED FOUND IN VERIFICATION

Valley Water's July 2025 plan already sets four stages for the Sunnyvale East and West Channels. Writing new ones would spend the weeks before the peak duplicating them.

* Suggested formulation: check with a health worker before sharing.

Instead: the city adopts and tests the existing stages (action 6).

Do not treat the East Channel upgrade as protection this winter

`CONTRAINDICATED` `HISTORIAN; CONFIRMED`

Completion is "anticipated by 2032"; this winter the 10-year channel that flooded in 1983 and 1998 is what exists.

Instead: the stages, clearing, and sandbags (actions 6 and 11).

Do not run a flood-insurance drive across the whole East Channel area

`CONTRAINDICATED` `FOUND IN VERIFICATION`

ZIP 94086 has had three flood claims since 1978, one of them paid ($8,854), and most of the 1,618 project parcels lie in neighboring ZIPs. Premiums would buy little protection here.

Instead: contents cover only for renters in the mapped East Evelyn area (action 10).

Do not send residents to PG&E's Portable Battery Program

`CONTRAINDICATED` `CARRIED; CORRECTED AND CONFIRMED`

"Applications for 2026 are closed." It also needs a past Public Safety Power Shutoff or three fast-trip outages, which few central Sunnyvale homes will have. Silicon Valley Clean Energy's battery pilot "is now complete."

Instead: Medical Baseline and supplier backup (action 5).

Do not count on a portable battery to keep an oxygen concentrator running*

`CONTRAINDICATED` `CARRIED; CONFIRMED`

A typical station gives about 1.7 usable kWh, three to six hours for a concentrator.

* Suggested formulation: check with a health worker before sharing.

Instead: the 27 home-oxygen users get backup cylinders from their supplier before December (action 5).

Do not tell people "you'll get an alert"

`CONTRAINDICATED` `CARRIED; CONFIRMED FOR SANTA CLARA COUNTY`

AlertSCC reaches cell phones only if they are registered.

Instead: register people at the door (action 7).

Do not tell residents to buy a generator*

`CONTRAINDICATED` `CARRIED; APPLIES HERE`

Portable generators cause more than 70 carbon-monoxide deaths a year in the US, and RVs and mobile homes fill quickly. In January 2023 two men died at Sea Ranch with one running in a hallway.

Instead: alarms and the indoor-fuel warning (actions 3 and 8).

Do not rely on the state's disaster utility protections before an emergency is declared

`CONTRAINDICATED` `CARRIED; CONFIRMED`

Under CPUC decision D.19-07-015 they apply only after the Governor or the President declares an emergency for the area.

Instead: CARE, FERA, and HEAP now (action 9).

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1915–1969	Groundwater pumping sinks the north valley by up to 13 feet	Runoff cut off from Stevens and Calabazas Creeks ponds in Sunnyvale	Imported water and recharge stop the sinking by about 1970; the channels are built
1963, 1968	Sunnyvale channel flooding	Not found	Not found
1982–83 (January 1983)	Strong El Niño: Calabazas Creek and both channels overtop	Homes and businesses damaged in Sunnyvale, Cupertino, and San José; deaths not found	Not found
1986	Channel flooding	One flood-insurance claim in 94086, $8,854	Not found
1997–98 (February 1998)	Strong El Niño: channel flooding; federal declaration DR-1203 includes the county	Not found for Sunnyvale	Not found
2015–16	Strong El Niño	Near-normal rain at Moffett Field; no channel flooding listed	—
February 2017	Coyote Creek flood, San José	About 14,000 ordered out	Joint San José–Valley Water creek emergency plan
2021–2023	Hypothermia deaths, Santa Clara County	42, up from 11 in 2018–2020	Not found
Winter 2022–23	Atmospheric rivers; federal declarations DR-4683 and DR-4699 include the county	Not broken down for Sunnyvale; channel overtopping not confirmed	—

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
July 2025	Valley Water emergency action plan with stages for the Sunnyvale channels	—	No city sign-off found
2026	Library warming center deactivated in a February storm; 72-hour RV rule enforced from 16 July	—	—

The recurrence nobody has noticed: both strong El Niño winters since the channels were built, 1982–83 and 1997–98, flooded them, and the same 10-year channel will be in the ground this winter. It has been 28 years since the last time, longer than the 20 years San José took to forget the lessons of 1997 before Coyote Creek flooded in 2017.

WHERE THE EXPERTS DISAGREE — RESOLVED AND OPEN

- **Resolved: flood insurance for the East Channel area.** The Emergency Manager wanted outreach to all 1,618 parcels; the Economist found three claims in 40 years. Verification sided with the Economist, except for renters where Valley Water's plan shows water reaching homes.
- **Resolved: are the warning triggers missing?** Two lenses proposed writing them. Verification found Valley Water already has them; what is missing is the city's adoption.
- **Open: rooms or kits?** The Philanthropist argued that past about 30 rooms, money should buy more nights rather than more rooms, because most vehicle dwellers will not leave their vehicles; verification cut the target to 20 rooms. The city's own $27,000 line, far below the pilot's budget, hints the rooms went under-used. Uptake data from the 2024–25 pilot would settle it, and was not found.
- **Open: is El Niño the right frame?** The premise test shows strong El Niño winters bring about 40% more rain here. The Public Health lens points out that the deaths come on ordinary cold, wet nights that no death certificate will attribute to El Niño. Both point to the same actions.

WHAT NOBODY COVERED — BLIND SPOT

No lens could measure this winter's deaths afterwards. Without a 2021–2026 baseline of winter deaths among unhoused and older residents, pulled now from the county Medical Examiner, nobody will know whether any of this worked. Nor could anyone confirm whether the council has

* Suggested formulation: check with a health worker before sharing.

already renewed the motel pilot for 2026–27; the city's agenda system could not be reached. A call to Sunnyvale's Housing Division would settle whether action 1 is a new decision or a renewal.

HOW TO READ THIS

METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighborhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The request gave ZIP code 94086, which was used as given: central Sunnyvale, Santa Clara County. Climate Impact Lab impact region: the one containing Sunnyvale, name not resolved. Mortality figures exist only at about 300,000 people; ZIP detail is used for hazards and actions, not death counts. Some figures are for the City of Sunnyvale or the county and are labelled as such.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (an emergency manager, a public health scientist, a frontline responder, a skeptic, an economist, a historian, and a philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief. They were given the verified state-level findings of the Belmont briefing (10 October 2026) as claims to re-check. The El Niño premise test was run on NOAA station data (Moffett Field USW00023244; San José USW00023293 as a cross-check) for this briefing.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by

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dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.

- **Verification.** Five verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not queried in this run: the inventory page shows no entries until a country is chosen in its interface, and the verifiers could not drive it. The Belmont briefing, run the same day, found no US entries covering backup power, carbon-monoxide alarms, or shelter for unhoused people. No action here is claimed to have, or to lack, a documented precedent.
- **Health wording.** Passages where a wording error could hurt someone (generators, vaccines, oxygen and medical equipment in an outage, and carbon monoxide) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. sunnyvale.ca.gov and the county's news site returned 403 to every request, and the city's agenda system was not reached; city facts marked as page extracts were cross-checked against local press. Unconfirmed: renewal of the 2024–25 motel pilot, the county's 2026–27 bad-weather thresholds, AlertSCC language support, channel overtopping in January 2023, and the Sunnyvale-specific toll of any past flood.

References · Premise: CPC Oceanic Niño Index · GHCN-D Moffett Field and San José (fetched) · CPC ENSO discussion (verified for Belmont, same day)
Unhoused residents: Sunnyvale 2025 count (page extract) · KFF Health News, hypothermia (verified) · SV Voice, winter motel pilot (verified) · WeHOPE contract (verified) · February 2025 motel placement (verified) · 72-hour RV rule (verified) · enforcement date (verified) · safe parking (verified) · Mountain View settlement (verified) · County bad-weather triggers, 2018–19 (verified, dated) · San José creek evacuations, 2023 (Skeptic) · CPSC generator deaths (verified)

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People and health: ACS 2024 5-year, ZCTA 94086 (verified) · HHS emPOWER (queried) · CDSS licensed facilities (queried) · Surie 2024, RSV (verified) · CDC RSV guidance (verified) · DiazGranados, NEJM 2014;371:635 (verified) · Yoon 1998 (verified) · HCD mobile-home CO rule (verified) · Anderson and Bell 2012 (verified)

Power, alerts, and money: PG&E DDAR fact sheet (verified) · Portable Battery Program (verified) · Medical Baseline (verified) · SVCE battery pilot (verified) · SVILC (verified) · AlertSCC FAQ (verified) · Sacred Heart utility assistance (verified) · FEMA flood insurance (verified) · CPUC disaster protections (verified)

Flood: Valley Water channel quick guide, 2025 (verified) · E2 channel project (verified) · Valley Water future projects, August 2026 (verified) · subsidence (verified) · OpenFEMA claims, 94086 (queried) · Sunnyvale sandbags (page extract) · SPUR on Coyote Creek · DR-1203, 1998 (verified)

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 10 October 2026 for ZIP 94086, Sunnyvale, California. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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